



## Solve with us 8 - Not Graded



This assignment will not be graded and is only for practice.

Key Points: 1Key Points: 1

- Two non-zero vectors  $u$  and  $v$  from an inner product space  $V$  are said to be orthogonal if  $\langle u, v \rangle = 0$ .
- A set  $S$  of non-zero vectors is said to be an orthogonal set of an inner product space if elements of the set  $S$  are mutually orthogonal. i.e.,  $\langle v_i, v_j \rangle = 0$  for  $i \neq j$ , any  $v_i, v_j \in S$ .
- Let  $\{v_1, v_2, \dots, v_n\}$  be an orthogonal set of vectors in an inner product space. Then  $\{v_1, v_2, \dots, v_n\}$  is a linearly independent set of vectors.
- Orthogonal basis: Let  $V$  be an inner product space. A basis consisting of mutually orthogonal vectors is called an orthogonal basis. In other word, a maximal orthogonal set is a basis and is called an orthogonal basis.
- An orthonormal set of vectors of an inner product space  $V$  is an orthogonal set of vectors such that the norm of each vector of the set is 1.
- An orthonormal basis is an orthonormal set of vectors which form a basis.
- Obtaining orthonormal set from orthogonal set: If  $\gamma = \{v_1, v_2, \dots, v_n\}$  is an orthogonal set of vectors then we can obtain an orthonormal set of vectors  $\beta$  from  $\gamma$  by dividing each vector by its norm i.e.,

$$\beta = \left\{ \frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}, \dots, \frac{v_n}{\|v_n\|} \right\}. \beta = \{ \|v_1\|^{-1} v_1, \|v_2\|^{-1} v_2, \dots, \|v_n\|^{-1} v_n \}.$$

Consider a set of vectors

$S = \{(1, -1, -1), (1, 2, 0), (-1, 1, 1), (-1, \frac{1}{2}, -\frac{3}{2}), (0, 3, 1), (2, 1, 1), (0, 1, -1)\}$  from the inner product space  $R^3$  with usual inner product (i.e., the dot product).

Answer questions (1) and (2).

1 point

Which of the following subsets of the set  $S$  consist of mutually orthogonal vectors?

☐

$\{(1, -1, -1), (1, 2, 0), (-1, 1, 1)\}$

☐

$\{(1, -1, -1), (-1, \frac{1}{2}, -\frac{3}{2})\}$

☐

$\{(1, -1, -1), (-1, \frac{1}{2}, -\frac{3}{2}), (1, 2, 0)\}$

☐

$S$

☐

$\{(2, 1, 1), (1, -1, -1), (0, 1, -1)\}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\{(1, -1, -1), (-1, \frac{1}{2}, -\frac{3}{2})\}$

$\{(2, 1, 1), (1, -1, -1), (0, 1, -1)\}$

1 point

Which of the following option(s) is (are) true?



☐  $\{(2, 1, 1), (0, 1, -1)\}$  is an orthogonal basis of a vector space which is isomorphic to  $R^2$ .

☐  $\{(2, 1, 1)\}$  is not an orthogonal basis of a vector space which is isomorphic to  $RR$ .

☐  $\{(1, -1, -1), (1, 2, 0)\}$  is an orthogonal basis of a vector space which is isomorphic to  $R^2$ .

☐  $\{(-1, 1, 1), (-1, \frac{1}{2}, -\frac{3}{2}), (0, 3, 1)\}$  is an orthogonal basis of  $R^3$ .

☐  $\{\frac{(2, 1, 1)}{\sqrt{6}}, \frac{(1, -1, -1)}{\sqrt{3}}, \frac{(0, 1, -1)}{\sqrt{2}}\}$  is an orthonormal basis of  $R^3$ .

☐  $\{(2, 1, 1), (1, -1, -1)\}$  is an orthonormal basis of  $R^3$ .

☐  $\{(1, -1, -1), (0, 1, -1)\}$  is an orthonormal basis of  $R^3$ .

☐  $\{(0, 1, -1)\}$  is an orthonormal basis of  $R^3$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\{(2, 1, 1), (0, 1, -1)\}$  is an orthogonal basis of a vector space which is isomorphic to  $R^2$ .

$\{\frac{(2, 1, 1)}{\sqrt{6}}, \frac{(1, -1, -1)}{\sqrt{3}}, \frac{(0, 1, -1)}{\sqrt{2}}\}$  is an orthonormal basis of  $R^3$ .

☐  $\{(2, 1, 1), (1, -1, -1)\}$  is an orthonormal basis of  $R^3$ .

☐  $\{(1, -1, -1), (0, 1, -1)\}$  is an orthonormal basis of  $R^3$ .

☐  $\{(0, 1, -1)\}$  is an orthonormal basis of  $R^3$ .

Key Points: 2

- The projection of a vector to a subspace: Let  $V$  be an inner product space,  $v \in V$  and  $W \subseteq V$  be a subspace. Then the projection of  $v$  onto  $W$  is the vector in  $W$ , denoted by  $proj_W(v)$ , computed as follows:

Find an orthonormal basis  $\{v_1, v_2, \dots, v_n\}$  for  $W$ . Then  $proj_W(v) = \sum_{i=1}^n \langle v, v_i \rangle v_i$ .

Remark:  $proj_W(v)$  is independent of the chosen orthonormal basis.

- Let  $V$  be an inner product space and  $W$  be a subspace. Then the projection of vectors in  $V$  to  $W$  is a linear transformation from  $V$  to  $W$  with image  $W$ .

1 point

Consider the vector space  $R^3$  with usual inner product. Which of the following vectors represents projection of the vector  $u = (2, 1, -2)$  onto the subspace whose an orthonormal basis is

$$\{(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}), (\frac{-2}{3}, \frac{1}{3}, \frac{2}{3})\} \{(\sqrt{2}, 0, \sqrt{2})\}$$

$$\sqrt{1, 0, \sqrt{2}}$$

$$\sqrt{1), (3-2, 31, 32)}\}$$

☐

$$(\frac{14}{9}, \frac{-7}{9}, \frac{-14}{9})(914, 9-7, 9-14)$$

☐ (1,1,1)

☐ (2,1,-2)

☐

$$(\sqrt{2}, 0, \sqrt{2})(\sqrt{2}, 0, \sqrt{2})$$

☐

$$, 0, \sqrt{2}$$

☐
☐

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$(\frac{14}{9}, \frac{-7}{9}, \frac{-14}{9})(914, 9-7, 9-14)$$

Consider the vector space  $R^3$  with usual inner product. If  $(a, b, c)$  is the projection vector obtained from projecting the vector  $(1, 0, 1)$  onto the subspace

$\{(x, y, z) \mid x - y + z = 0, x, y, z \in R\}$ , then find the value  $3(a + b + c)$ .

Hint: Find any orthonormal basis of the subspace  $\{(x, y, z) \mid x - y + z = 0, x, y, z \in R\}$  and use the definition of the projection of a vector onto a subspace.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 4

1 point

Key Points: 3

- Gram-schmidt process: The Gram-schmidt process is used to get an orthogonal basis from any given basis of a vector space. A slight modification yields an orthonormal basis.
- Let  $\gamma = \{v_1, v_2, \dots, v_n\}$  be a given ordered basis of a vector space. The steps in the process are described below and yield an orthonormal basis  $\beta = \{u_1, u_2, \dots, u_n\}$ .

Step-1  $w_1 = v_1$ , and  $u_1 = \frac{w_1}{\|w_1\|}$ .

Step-2  $w_2 = v_2 - \langle v_2, u_1 \rangle u_1$  and  $u_2 = \frac{w_2}{\|w_2\|}$ .

Step-3  $w_3 = v_3 - \langle v_3, u_1 \rangle u_1 - \langle v_3, u_2 \rangle u_2$  and  $u_3 = \frac{w_3}{\|w_3\|}$ .

$$\vdots$$

Step-i  $w_i = v_i - \sum_{j=1}^{i-1} \langle v_i, u_j \rangle u_j$  and  $u_i = \frac{w_i}{\|w_i\|}$ .

$$\vdots$$

Step-n  $w_n = v_n - \sum_{j=1}^{n-1} \langle v_n, u_j \rangle u_j$  and  $u_n = \frac{w_n}{\|w_n\|}$ .

Let  $\beta$  be an ordered basis of the inner product space  $R^3$  with usual inner product, where  $\beta = \{(1, 1, -1), (1, 1, 0), (1, 0, 0)\}$  and  $\gamma = \{u_1, u_2, u_3\}$  be the corresponding orthonormal basis obtained using the Gram-schmidt process. Use this information to answer the questions (5), (6) and (7).

If  $u_1 = (a_1, b_1, c_1)$ , then find the value of  $\sqrt{3}(a_1 + b_1 + c_1)$ .

$$\sqrt{3}$$

(a1 + b1 + c1).

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

If  $u_2 = (a_2, b_2, c_2)$ , then find the value of  $\sqrt{6}(a_2 + b_2 + c_2)$ .

$$\sqrt{6}$$

(a2 + b2 + c2).

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 4

1 point

If  $u_3 = (a_3, b_3, c_3)$ , then find the value of  $\frac{1}{\sqrt{2}}(a_3 + b_3 + c_3)$ .

$$\frac{1}{\sqrt{2}}$$

1(a3 + b3 + c3).

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 0

1 point

Key Points: 6

- Orthogonal transformation: Let  $V$  be an inner product space and  $T$  be a linear transformation from  $V$  to  $V$ .  $T$  is said to be an orthogonal transformation if

$$\langle Tv, Tw \rangle = \langle v, w \rangle, \forall v, w \in V$$

- Orthogonal matrix: Orthogonal matrix: A square matrix  $A$  is called an orthogonal matrix if

$$AA^T = A^T A = I \quad AA^T = A^T A = I \text{ (identity matrix)}$$

1 point

Consider the following linear transformations  $T_1, T_2: R^2 \rightarrow R^2$ , and  $T_3, T_4: R^3 \rightarrow R^2$   $T_1, T_2: R^2 \rightarrow R^2$ , and  $T_3, T_4: R^3 \rightarrow R^2$  defined as

$$T_1(x, y) = \left(\frac{x+y}{\sqrt{2}}, \frac{x-y}{\sqrt{2}}\right), T_1(x, y) = \left(\frac{x+y}{\sqrt{2}}, \frac{x-y}{\sqrt{2}}\right),$$

$$\sqrt{\frac{x+y}{2}}, \sqrt{\frac{x-y}{2}},$$

$$T_2(x, y) = (x + 3y, 2x - 2y), T_2(x, y) = (x + 3y, 2x - 2y),$$

$$T_3(x, y, z) = \left(\frac{x+z}{\sqrt{2}}, \frac{x+y\sqrt{3}+z}{\sqrt{3}}\right), T_3(x, y, z) = \left(\frac{x+z}{\sqrt{2}}, \frac{x+y\sqrt{3}+z}{\sqrt{3}}\right),$$

$$\sqrt{\frac{x+z}{3}}, \sqrt{\frac{x+y}{3}}, \sqrt{\frac{x+y}{3}},$$

$$T_4(x, y, z) = (x + y + z, x + y), T_4(x, y, z) = (x + y + z, x + y),$$

where  $R^3$   $R^3$  and  $R^2$   $R^2$  are the inner product spaces with respect to the dot product. Which of the linear transformation is an orthogonal transformation?

☐  $T_1$   $T_1$

☐  $T_2$   $T_2$

☐  $T_3$   $T_3$

☐  $T_4$   $T_4$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T_1$   $T_1$

1 point

Which of the following option(s) is (are) true?

☐

$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$  is an orthogonal matrix.

☐

$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$  is an orthogonal matrix.

☐

$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$  is an orthogonal matrix.

☐

$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 2 \\ -1 & -2 & 1 \end{bmatrix}$  is an orthogonal matrix

☐

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

$\sqrt{\phantom{x}}$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

$\begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$  is an orthogonal matrix.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$  is an orthogonal matrix.

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$\sqrt{\phantom{x}}$

$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$  is an orthogonal matrix.

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

$\sqrt{\phantom{x}}$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

$\begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$  is an orthogonal matrix.